

ASSIGNMENT No: 4

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TITLE: Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Problem Statement: Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Learning Objectives: To perform conversion between primitive data types and wrapper objects using Java wrapper classes.

Theory: Wrapper classes are predefined classes that encapsulate primitive data types into objects. Java provides wrapper classes such as Integer, Double, Character, and Boolean for every primitive type. They support auto boxing, unboxing, parsing, and conversion methods. Wrapper classes are widely used in Java Collections Framework and when object representation of primitive values is required.

Exercise

1. Create a program to convert student marks from String format to Integer and calculate total marks.

```
StudentMarks.java > StudentMarks > main(String[])
1  public class StudentMarks {
2      Run | Debug
   public static void main(String[] args) {
3
4       String mark1 = "85";
5       String mark2 = "90";
6       String mark3 = "78";
7
8       // Convert String to Integer
9       int m1 = Integer.parseInt(mark1);
10      int m2 = Integer.parseInt(mark2);
11      int m3 = Integer.parseInt(mark3);
12
13      int total = m1 + m2 + m3;
14
15      System.out.println("Marks 1: " + m1);
16      System.out.println("Marks 2: " + m2);
17      System.out.println("Marks 3: " + m3);
18      System.out.println("Total Marks: " + total);
19  }
20  }
21

PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\arhaa\OneDrive\Desktop\java> & 'C:\Program Files'
ng\Code\User\workspaceStorage\02e7f2cfce92a0ba8e30606ade6b2c24
Marks 1: 85
Marks 2: 90
Marks 3: 78
Total Marks: 253
PS C:\Users\arhaa\OneDrive\Desktop\java>
```

2. Develop an Employee Payroll System that accepts employee IDs, basic salary, and bonus amounts from the user. Convert the entered values into wrapper objects and perform validation operations to ensure valid salary values before calculating the net salary.

```
EmployeePayroll.java
1 import java.util.Scanner;
2 public class EmployeePayroll {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5
6         System.out.print("Enter Employee ID: ");
7         String idInput = sc.nextLine();
8
9         System.out.print("Enter Basic Salary: ");
10        String salaryInput = sc.nextLine();
11
12        System.out.print("Enter Bonus Amount: ");
13        String bonusInput = sc.nextLine();
14
15        // Convert to Wrapper Objects
16        Integer empId = Integer.valueOf(idInput);
17        Double basicSalary = Double.valueOf(salaryInput);
18        Double bonus = Double.valueOf(bonusInput);
19
20        // Validation
21        if (basicSalary < 0 || bonus < 0) {
22            System.out.println("Invalid Salary or Bonus Amount!");
23        } else {
24            double netSalary = basicSalary + bonus;
25
26            System.out.println("\nEmployee ID: " + empId);
27            System.out.println("Basic Salary: " + basicSalary);
28            System.out.println("Bonus: " + bonus);
29            System.out.println("Net Salary: " + netSalary);
30        }
31
32        sc.close();
33    }
34 }
35 }
```

```
PS C:\Users\arhaa\OneDrive\Desktop\java> &
ng\Code\User\workspaceStorage\02e7f2cfce92a
Enter Employee ID: 131
Enter Basic Salary: 5600
Enter Bonus Amount: 600

Employee ID: 131
Basic Salary: 5600.0
Bonus: 600.0
Net Salary: 6200.0
PS C:\Users\arhaa\OneDrive\Desktop\java>
```